

Banking Transactions & Customer Financial Behavior Analysis

A data-driven framework using **Python**, **SQL**, and **Power BI** to decode customer financial behavior, reduce risk, and unlock revenue.

Python

Data cleaning, EDA, feature engineering

SQL

Aggregated metrics & decision queries

Power BI

Executive & operational dashboards

5

Business Challenges

Addressed end-to-end

6

Workflow Stages

From raw data to decisions

6

Dashboard Views

Executive & operational

6

Recommendations

Actionable next steps

The Business Problem: Why This Analysis Matters

Decision-makers face five interconnected challenges that erode profitability and elevate risk. This project delivers a unified data-driven framework to address all five.

01 — Fraud Exposure

Undetected fraudulent transactions erode customer trust and generate direct financial losses. Early detection — powered by behavioral pattern analysis — is not optional; it is a competitive baseline requirement.

02 — Revenue Concentration Risk

A small segment of high-value customers drives a disproportionate share of total revenue. Without a structured retention program, the loss of even a handful of these customers has outsized financial consequences.

03 — Loan Decision Inefficiency

Inconsistent approval criteria create a dual risk: missed revenue from creditworthy applicants who are declined, and elevated default exposure from approvals made without data-backed thresholds.

04 — Branch Performance Gaps

Significant variance in branch-level revenue signals untapped optimization potential. Top-performing branches outperform the median by a material margin — and the drivers of that outperformance are replicable.

05 — Channel Underutilization

Misalignment between customer channel preferences and bank investment priorities reduces engagement and pushes customers toward competitors with stronger digital offerings. Channel analytics close this gap.

📄 This project delivers a data-driven framework to address all five challenges using Python, SQL, and Power BI — producing both executive-level visibility and operational decision support.

Project Workflow: From Raw Data to Business Decisions

Each stage feeds directly into the next – ensuring that every insight delivered to decision-makers is grounded in clean, validated, and reproducibly queried data.



The pipeline is end-to-end: stakeholder KPIs defined in Stage 1 are the same metrics surfaced in the Power BI dashboard in Stage 5 and translated into recommendations in Stage 6. There is no disconnect between what the business needs and what the analysis delivers.

Upstream (Stages 1–3)

- **Business Understanding:** Define KPIs, success metrics, and stakeholder requirements before touching the data
- **Data Cleaning (Python):** Handle missing values, remove duplicates, normalize formats, and flag anomalies
- **EDA:** Uncover distributions, correlations, and behavioral patterns across customer segments

Downstream (Stages 4–6)

- **SQL Queries:** Extract aggregated metrics – transaction volumes, loan approval rates, branch revenue, channel usage
- **Power BI Dashboard:** Interactive visualizations designed for executive and operational reporting layers
- **Business Insights:** Findings translated into actionable recommendations for each business unit

Data Cleaning & Preparation (Python)

Analytical quality is determined before the first SQL query is written. A rigorous six-step cleaning protocol ensures that every downstream metric is derived from validated, trustworthy data.



Missing Values

Imputation strategies applied per column type – **median** for numeric fields, **mode** for categorical. No blanket deletion that would bias distributions.

Deduplication

Transaction-level deduplication to prevent inflated volume metrics. Composite keys (customer ID + timestamp + amount) used as uniqueness constraints.

Outlier Detection

IQR and **Z-score** methods flag anomalous transaction amounts for downstream fraud review – preserving signal without distorting aggregates.

Feature Engineering

Derived fields: transaction frequency per customer, average ticket size, **credit score bands**, and channel preference scores – all critical for segmentation.

Type Normalization

Date parsing, currency standardization, and categorical encoding applied for full **SQL and Power BI compatibility** – no manual transformation at query time.

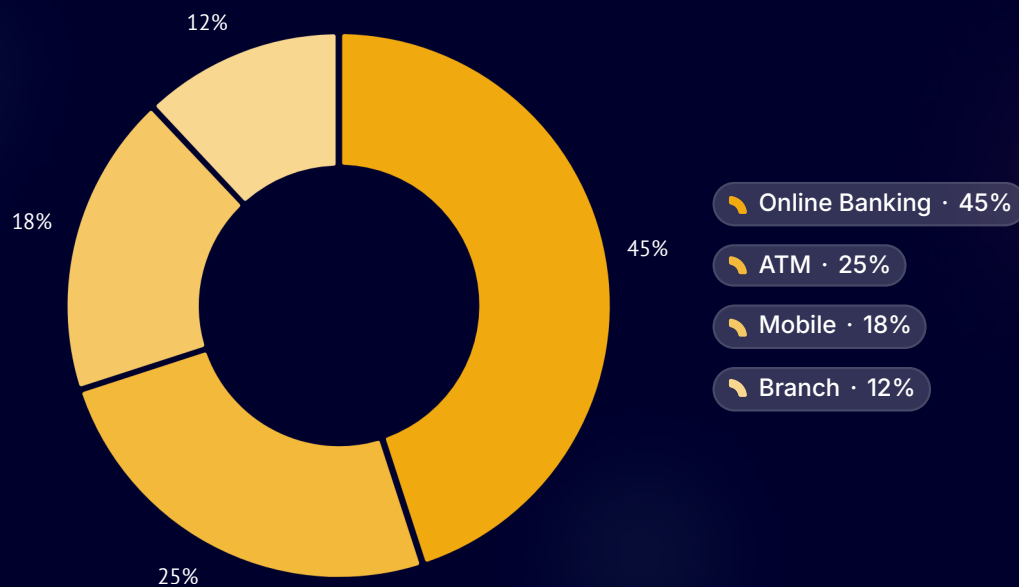
Validation Checks

Post-cleaning assertions: row counts, null rates, and value range checks confirm data integrity before any analysis proceeds.

Exploratory Data Analysis: Key Patterns Uncovered

EDA surfaces the behavioral and structural patterns that define where risk concentrates, where revenue is generated, and where operational gaps exist across the customer base.

Transaction Channel Distribution



Online banking dominates at **45%** — far ahead of ATM (25%), Mobile (18%), and Branch (12%). Digital investment must reflect this reality.

Additional EDA Findings

Pareto Revenue Effect

A small percentage of customers contribute the majority of total revenue — a classic 80/20 distribution confirming the criticality of high-value customer retention programs.

Branch Performance Variance

Top-performing branches generate substantially higher revenue than the median — driven by customer mix and transaction volume, not geography alone.

Fraud Signal Clustering

Anomalous transactions cluster around specific time windows, geographies, and unusually high ticket sizes — providing a rule-based foundation for real-time flagging.

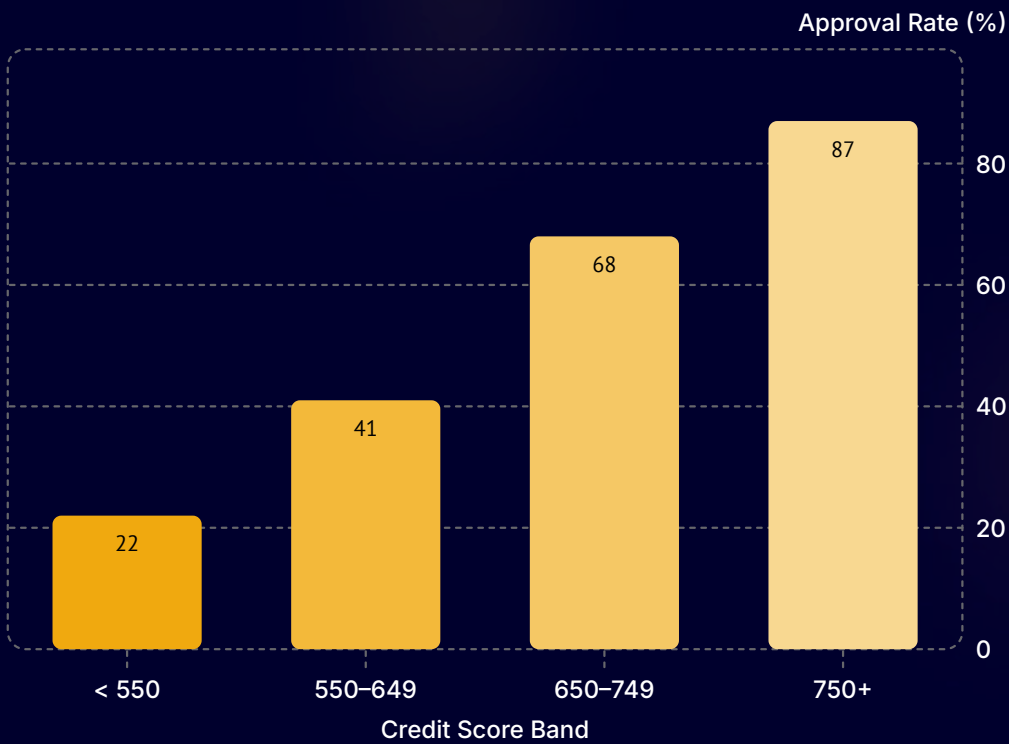
Customer Segmentation

Behavioral clustering reveals two dominant profiles: **high-frequency low-value** and **low-frequency high-value** — each requiring a distinct product and engagement strategy.

SQL Analysis: Metrics That Drive Decisions

Six purpose-built SQL queries translate cleaned transaction data into the decision-ready metrics that underpin every business recommendation in this project.

Loan Approval Rate by Credit Score Band



Approval rates rise sharply above the **650 threshold** – from 41% to 68% – confirming credit score banding as the most reliable primary filter for automated decisioning.

Query Inventory & Business Purpose

01

Branch Revenue Ranking

Aggregate revenue per branch identifies top performers and underperformers – enabling targeted resource reallocation decisions.

02

High-Value Customer Identification

Customers ranked by total transaction value – the top decile is flagged automatically for retention and relationship management programs.

03

Fraud Flagging Logic

Transactions exceeding **3× the customer's average ticket size**, or originating from new geographies, are surfaced immediately for compliance review.

04

Channel Preference by Segment

Cross-tabulation reveals which segments are underserved digitally – directing product and UX investment toward highest-impact cohorts.

05

Monthly Transaction Trend

Time-series aggregation tracks volume and value trends over rolling periods to support revenue forecasting and capacity planning.

Power BI Dashboard: Turning Data into Visibility

The dashboard is architected for two distinct audiences: **executive oversight** requiring high-level KPIs, and **operational teams** requiring drill-down capability to the transaction level. Six dedicated views serve both layers simultaneously.

1	Executive Summary KPI cards for total revenue, transaction volume, active customers, and loan approval rate – refreshed in real time. Designed for C-suite consumption in under 60 seconds.
2	Branch Performance Geographic map combined with a bar chart ranking branches by revenue and transaction count. Instantly surfaces top performers and outliers for regional leadership.
3	Customer Segmentation Scatter plot of transaction frequency vs. value, color-coded by customer tier. Enables relationship managers to identify high-value customers at risk of churn.
4	Fraud Monitoring Flagged transactions table with live filters by date, branch, and risk score. Compliance teams can triage anomalies without exporting to separate tools.
5	Channel Analytics Trend lines showing channel usage over rolling time periods – provides the quantitative basis for digital investment prioritization decisions.
6	Loan Analysis Approval rate breakdown by credit score band, loan type, and branch. Operationalizes the SQL findings into a self-service decision support tool for credit teams.

📌 The dashboard eliminates the reporting lag between data availability and decision-making – enabling every business unit to act on current intelligence, not last month's exports.

Business Recommendations: What the Bank Should Do Next

Six concrete, evidence-backed recommendations – each directly traceable to a finding from the EDA, SQL analysis, or dashboard – translate this project's outputs into measurable business action.

01 — Double Down on Digital

Online banking leads all channels at **45% of usage**. Prioritize UX investment, feature expansion, and fraud controls specifically for the online platform. Mobile at 18% represents the fastest-growth opportunity.

02 — High-Value Retention Program

The top revenue-contributing segment – identified via transaction value ranking – warrants dedicated relationship management, personalized product offers, and proactive outreach before attrition signals emerge.

03 — Automate Loan Decisioning

Use credit score banding as the primary filter: **automate approvals above 750** (87% historical approval rate), flag **650–749** for expedited review (68%), and reduce costly manual processing for clear-cut cases.

04 — Replicate Top Branch Practices

Identify the specific drivers of top-branch outperformance – customer mix, staff training, and product mix – and build a structured playbook for application across underperforming branches.

05 — Real-Time Fraud Detection

Integrate the SQL-based flagging logic (transactions exceeding **3× average ticket size** or from new geographies) directly into the transaction processing pipeline for immediate alerts – not batch-based review.

06 — Segment-Driven Marketing

High-frequency customers → loyalty and rewards products. **High-value low-frequency customers** → premium services and private banking pathways. Generic campaigns leave revenue on the table.

Each recommendation is supported by a reproducible query or visualization in the Power BI dashboard – enabling ongoing tracking of implementation impact.